INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 6: Conclusion and Future Work

by

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6 Conclusion and Future Work

Chapter 5 interpreted the findings and judged whether the study met its objectives. The tables of Chapter 4 form the record of the revised analysis, and Table 8.13 in Appendix 8.6.10 keeps the decision contrasts and verdicts of the registered analysis.

6.1 Answers to the research questions

The thesis proposed EndorseRank, PageRank with the edge weights of Adaptive Weighted PageRank (AWP) on a latest-allowance endorsement graph, and evaluated it beside AWP itself (T. Do et al., 2023), a coupled walk over both edge types and the raw degrees these walks smooth. Every scored address was a smart contract on Arbitrum One. Both graphs kept the approvals and transfers of WETH, USDC, USDT0, ARB and WBTC, with every amount valued in US dollars. In the same window the scores ranked the matched cohort, the contracts approved on these tokens by at least three distinct owners between October 2023 and June 2026 (Section 4.1.1). Out of window they ranked the contract spenders that held a positive allowance of these tokens at a freeze date, with labels counted from April to June 2026 (W0) and from July to September 2026 (W1); the rule for W1 was registered before the study extracted its logs (Sections 3.7 and 3.7.2).

The amounts were revised on 10 October 2026, after the results of the registered analysis were known (Section 3.5.1). The analysis plan had fixed amounts in raw token base units, every ERC-20 token and a value weight that was close to one for almost any amount. The revision values every amount in US dollars, keeps the five tokens with the most rows among those that pass a price-feed rule and have a daily price, sets the slope of the value weight so that the median amount weighs one half and caps the allowances of the coupled walk; it was committed before any revised score was computed. The answers below come from the revised analysis. Rules A and B and the

comparison on traders keep the thresholds and the wording fixed for the registered analysis, but they were applied to the revised scores after the registered results were known, so their verdicts here are a re-analysis and not registered tests. The registered analysis reached the same verdicts (Appendix 8.6.10; Table 8.13).

RQ1 asked whether EndorseRank and AWP order the contracts of one cohort differently. They do, and H1 is supported, largely by construction (Section 4.4; Claim C3). Every cohort contract received approvals of one of the five tokens from at least three owners, so EndorseRank leaves few pairs of contracts tied, and keeping the contracts that a graph misses as isolated nodes leaves the coefficient between the two scores unchanged (Table 4.5). The divergence therefore comes from the two orderings themselves. Whether they reflect different kinds of reputation is a question for the checks of RQ2 and the holdouts of RQ4, not for the divergence alone.

RQ2 asked how closely each score agrees with allowance and transfer checks in the same window. Each score agrees most with the proxies of the edge type it walks on, EndorseRank with the allowance family and AWP with the transfer family, and the paired contrasts between families and between the two scores, none of them fixed in advance, have intervals that exclude zero; the C-PR mixtures lie between their two layers on every family (Tables 4.3 and 8.7; Claim C2). With amounts in US dollars, AWP follows the summed value of a contract's inbound transfers almost as closely as the number of its senders. In the narrow sense of agreement with the checks of a score's own edges, H2 is supported, largely by construction. The proxies are built from the same events and amounts as the graphs, so the agreement is an intended-construct check, and part of it is built in.

RQ3 asked what each score costs to compute on the same contracts. On the full graphs of the matched cohort, the time of a solver call follows the edge count of its graph. The allowance graph has less than half as many edges as the transfer graph, EndorseRank's call takes correspondingly less time than AWP's, and C-PR and S-PR, which also walk the transfer layer, take longer than AWP (Section 4.2; Claim C1). H3 is supported, largely by construction. The same solver runs on every graph, so the speed belongs to the sparser graph and not to the algorithm.

RQ4 asked whether scores frozen at a given date predict the new approval pairs and new transfer

senders that spender contracts gain afterwards, and whether any PageRank predicts them better than the raw degree it smooths. The first part holds in both windows: the interval of every score excludes zero on both labels, and in contrasts computed after the labels EndorseRank ranks later approval pairs above AWP, while AWP ranks later senders above EndorseRank (Tables 4.10, 4.14, 4.16 and 8.11). The second part has a different answer for each walk (Claim C4). AWP ranks later new senders above the raw transfer in-degree in both windows, so on the transfer side the walk, with its time and value weights and activity restarts, added information that the count lacks. EndorseRank with uniform restarts ranks later approval pairs below the raw in-approve degree in both windows (Tables 4.11 and 4.16). Leaving out the spenders that had received no transfer, which AWP scores zero, keeps both degree comparisons of W0 in place (Table 4.12). These degree contrasts were computed after their labels were known, but each points the same way in both windows, and the registered analysis gave the same directions (Table 8.13). Multi-hop propagation is present in these graphs: some spenders are approved by owners that are themselves approved, and for most of them EndorseRank departs from its depth-one closed form. It still did not lift EndorseRank above the count (Section 5.3.2). With AWP's activity restarts, EndorseRank ranks later approval pairs above the degree in both windows (Section 4.6; Tables 4.9 and 4.16), but that comparison lies outside every rule, in the registered analysis as in the revised one, and is a hypothesis for a registered test, not a finding.

RQ5 asked whether a coupled PageRank over both edge types improves on a walk over the transfers alone, and whether it ranks contracts better than a walk over either layer alone. On the revised scores both answers are no, as they were in the registered analysis (Claim C5). Rule A, written in the analysis plan before any log of the study was extracted, asked C-PR to do no worse than the allowance walk on new approval pairs and better than the transfer walk on new transfer senders in W0. The first condition holds and the second fails, and the same-window part of the rule fails as well (Section 4.3.1; Tables 4.11 and 4.4). Rule B, registered before the study extracted the logs of W1, asked C-PR to beat the transfer walk on new approval pairs and to be no worse than it, within a fixed margin, on new transfer senders and on the traders' liquidation-free close rate. The first condition

holds; it also held in the registered analysis, where it tested whether the gain seen in W0 on new approval pairs would replicate. The loss on new senders exceeds the margin. On the trader label the two scores have equal point estimates, but the interval of their difference reaches further below zero than the margin allows, so that condition fails narrowly; with AWP in place of the transfer walk, the first sensitivity analysis of the registration, the loss on new senders is larger still (Table 4.15). In contrasts computed after the labels, the seeded ablation S-PR ranks new senders almost as the transfer walk does and ranks new approval pairs below the allowance walk (Table 4.11). Allowance edges help when they carry the walk, not when they only decide where it restarts.

Beside the research questions, the registration of W1 set allowance-side against transfer-side scores on an outcome built from neither edge type, the liquidation-free close rate of GMX V2 contract traders (Section 4.6.3; Claim C6). On the revised scores, and in the wording fixed at registration, there is no count-level lead of allowance over transfer on later liquidation avoidance: the approval count led the transfer count in W0, but in W1, the primary window, the interval of the contrast includes zero. The PageRank versions are not resolved. In W1 the approval count ranks traders below the transfer count on the profitable share and on the non-loss share, so whatever signal it carries concerns forced-liquidation avoidance, not trading success (Table 8.12). Every trader that had received an approval of the five tokens avoided liquidation in both windows, and some of the other traders did not (Table 4.19); the transfer count, however, also ranks this label in W1 (Table 4.14). Contract accounts traded little on GMX V2 in 2026, so these samples are small and every trader interval is wide.

Claims C1–C6 (Section 4.7) state these findings formally.

The main research question asked what ERC-20 allowance edges add to PageRank as a reputation score for smart contracts on Arbitrum One, compared with the transfer walk and with the raw degrees that both walks smooth. Compared with the transfer walk, the allowance edges of the five tokens, valued in US dollars, add a second construct. The allowance walk orders contracts by the spending authority that owners delegate to them, each allowance weighted by its dollar value through a bounded weight; this ordering differs from the ordering by transfers, agrees with the

allowance checks and is computed on a sparser graph. Allowance-based scores also rank later approval pairs above the transfer walk: C-PR beat its transfer layer on that label in W1 under the first condition of rule B, and EndorseRank beat AWP in both windows in contrasts computed after the labels (Tables 4.15 and 8.11). Compared with the raw degrees, the allowance walk with uniform restarts did worse than the count it smooths. In contrasts computed after the labels, EndorseRank ranked later approval pairs below the count of approving owners in both windows, whereas AWP improved on its own count. Mixed into one walk with the transfers, allowance edges lift the ranking of later approval pairs above that of either layer walked alone, by a small margin over the allowance layer, but they lower the ranking of later senders below that of the transfer walk, so the mixture does not improve on the better single layer for both labels at once. On the one outcome built from neither edge type, the rules fixed at registration found no lead for the allowance side, on the revised scores as in the registered analysis.

6.2 Original contribution to knowledge

In the terms of the literature reviewed in Chapter 2, the five contributions of Section 1.9 establish the following.

Before this work, graph reputation on public ledgers ranked accounts by their transactions. H.-D. Do and Do (2023) applied PageRank and HodgeRank to Ethereum transaction graphs, AWP weighted transfer edges by value and age (T. Do et al., 2023), and RiskProp propagated risk through the transaction network (D. Lin et al., 2025). Blockchain reputation systems mostly took explicit ratings or transfer volume as their trust signal (Bellini et al., 2020), the attestation tooling deployed in DeFi scores the holder of an address from credentials (Human Passport, n.d.), and the risk of a contract has been judged from its code (Z. Lin et al., 2024). None of this work used the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power (Vogelsteller & Buterin, 2015), as the edge of a reputation ranking, and none ranked contracts by the approvals they receive. Approvals had been used to cluster the addresses of one entity (Victor, 2020) and measured as a security risk

(Wang et al., 2022). Nor had on-chain reputation scores been evaluated on one cohort against the checks of each construct and against the raw degrees they smooth, with an interval for every coefficient, paired contrasts and temporal holdouts judged by rules committed before the evaluation extracted its data. This thesis supplies both, and its main results apply those rules to amounts revised after the registered results were known (Section 3.5.1). PageRank (Page et al., 1999) and AWP's weighting are taken from the literature, with the slope of the value weight set here from the median dollar amount; EndorseRank changes only the edge, and the coupled operator changes the walk.

The new knowledge has five parts. First, what an on-chain reputation score of a contract measures depends on the edges it walks, and this can be checked on one cohort with quantified uncertainty: the allowance walk and the transfer walk order the same contracts differently, and each agrees most with the checks of its own edges (Claims C2 and C3). The allowance graph is also the sparser of the two, so its walk costs less; that follows from its edge count and is not a property of the method (Claim C1). Second, out of window the two walks relate differently to the counts they smooth, in contrasts computed after the labels were known. The transfer walk, AWP, ranked later new senders above the transfer in-degree in both windows. The allowance walk, EndorseRank with uniform restarts, ranked later approval pairs below the in-approve degree in both windows (Claim C4), even though the allowance-based scores ranked those pairs above the transfer walk (Tables 4.15 and 8.11). Third, the closed form accounts for part of this result. For a spender whose owners receive no approvals, EndorseRank is a linear function of the owner-normalized in-strength, which divides each owner's endorsement over all the spenders it approves. For the other spenders the walk propagates over more than one step, so multi-hop propagation is present in these graphs; over all spenders EndorseRank still did not beat the count, and whether propagation adds anything where it occurs was not tested separately (Section 5.3.2). A propagated allowance score therefore has yet to earn its place against the count, and the variant with activity restarts is the candidate for that test. Fourth, a weighted combination of the two edge types does not improve on the better single layer for both spender labels at once: it gains a little over the allowance walk on later approval pairs and loses against the transfer walk on later senders. Under rule B it beats the transfer walk

on later approval pairs, but its loss on later senders exceeds the margin and it cannot show that it is no worse on the traders' label, so the rule is not met (Claim C5). Fifth, on an outcome that neither graph records, the comparison registered for W1 found no count-level lead of allowance over transfer and left the PageRank pair unresolved (Claim C6). On the few contract traders available, the comparison did not separate allowance-edge from transfer-edge information, and this is not evidence that allowance edges carry no such information. The fourth and fifth parts apply rules fixed at registration to the revised scores after the registered results were known, so their verdicts are a re-analysis; the registered analysis reached the same verdicts (Appendix 8.6.10).

The construct-validity framing (Cronbach & Meehl, 1955) that makes these claims testable is part of the contribution. So is the design that fixed the cohort, the scores and the decision rules before the study extracted the logs they were judged on (Section 3.7). The registered analysis followed it; the revision of the amounts departed from it after the registered results were known, but was committed before any revised score was computed and is reported beside the registered verdicts (Appendix 8.6.10). The open pipeline publishes the summaries of both analyses, so others can check every table (Section 3.13), since reputation systems for pseudonymous users must remain auditable (Packin & Lev-Aretz, 2024).

The scope is deliberately narrow. DeFi lending rests on collateral (Schär, 2021), and the thesis does not show that on-chain reputation can replace it. It does not show that the results extend beyond contracts approved on WETH, USDC, USDT0, ARB or WBTC by at least three owners on Arbitrum One in the months studied, or to the approvals and transfers of other tokens, and it does not show that any score measures the safety of a contract, the creditworthiness of an account or trading skill (Section 2.8). Nor does it claim that EndorseRank resists Sybils. Under uniform teleportation a small closed farm of owner addresses makes a contract look ordinary or better at little cost, although reaching the top of the cohort would take a farm larger by orders of magnitude. Keeping only the five tokens does not raise that cost, since approving a token needs no balance of it (Section 5.6).

6.3 Future work

Each suggestion below answers a limitation discussed in Chapter 5. Any new score, weighting or label it leads to should be fixed and registered before the logs it is judged on are extracted for it, as the decision rules of the registered analysis were and the revised amounts were not, because registration keeps a confirmatory test apart from exploration (Nosek et al., 2018). It should also be checked out of window against the raw degrees.

- A registered test of the revised analysis. The US-dollar amounts, the five tokens, the value slopes and the allowance cap of the main analysis were fixed after the registered results were known (Section 3.5.1), so its verdicts are a re-analysis even where they agree with the registered ones (Table 8.13). A registration committed before the logs of a later window are extracted, such as October to December 2026, would fix these choices together with the cohort rule, rules A and B, the margin and the comparison on traders, and would test whether the verdicts of the revised analysis hold on logs extracted after the registration. The value-slope check moved the same-window family means little (Table 4.7), but it did not cover the holdout labels, so the slopes and the cap should be fixed in the registration and not tuned on the new window.
- A registered test of EndorseRank with activity restarts against the in-approve degree. With AWP's restarts, which weight each owner by how actively it approves, EndorseRank ranked later approval pairs above the raw degree in both windows, but the comparison was made after the labels were known (Section 4.6; Table 4.16). A test on a later window whose logs have not been extracted, which could form part of the registration above, would fix the contrast, its margin and the label in advance and keep the uniform-restart version as a second comparator. It should report the contrast separately for spenders whose owners receive no approvals, where the walk reduces to a weighted count of approving owners, and for the other spenders, where it propagates over more than one step (Section 5.3.2). Only the second group can show whether propagation, and not the restart weights alone, adds to the count.

- Seed-restricted teleportation against farms. Under uniform teleportation every address, a fresh one included, receives the same restart mass, so a closed farm of owner addresses collects more than its share of the population and a small farm lifts the contract it approves (Section 5.6). Restricting teleportation and the redistribution of dangling mass to a vetted seed set, as TrustRank does (Gyöngyi et al., 2004), would remove that inflow, and some asymmetric reputation functions based on flows from a trusted node resist Sybil attacks where PageRank does not (Cheng & Friedman, 2005). Such a variant should be evaluated under the attacks of the cost model and on the same holdout labels, with attention to which contracts the choice of seeds favours. The cost model is analytical and no attack was carried out on chain, so an empirical study with wash approvals and coordinated owner farms is needed to measure what an attack costs in practice.
- Combination with personhood credentials. EndorseRank uses no off-chain information, so it cannot tell whether the owners that approve a contract are distinct people. Proof of personhood does not apply to a contract, but it does apply to the externally owned accounts among its owners. Counting only the approvals of owners that pass a Passport-style check (Human Passport, n.d.) would make every farm address cost a credential as well as gas. The weaknesses of the gate, such as credential farming (Bordeianu & Popescu, 2026), would carry over, and personhood protocols trade Sybil resistance against privacy (Siddarth et al., 2020). Evaluating the combination needs labelled Sybil data, which this study did not collect, and a rule for owners that are themselves contracts.
- Labelled contract-risk outcomes. The out-of-window labels of this thesis count later approval pairs and senders, and the trader label covers few contracts; none of them says whether a contract is safe. The revocation and drain counts computed on the holdout rose with every score and with the raw degrees, so they track volume and not risk (Section 4.6). A test against labelled outcomes, such as contracts later tied to drainer phishing operations (He et al., 2025) or flagged for contract-related rug-pull risk (Z. Lin et al., 2024), with scores frozen before the label window and compared within strata of activity, would show whether the standing of

a contract among its owners says anything about its risk beyond its volume. Such labels are rare, so the window and the cohort would need to be larger than here.

- Other chains and later windows. The results hold for contracts approved on one of the five tokens by at least three owners on Arbitrum One in the months studied. Contracts on Arbitrum face security risks that differ from those on Ethereum mainnet (Tang & Shi, 2024), and DeFi protocols differ widely in design (Werner et al., 2022), so the same analysis is needed on other rollups, on Layer-1 networks and over later windows before any result is read as general. A chain or protocol on which more contract accounts trade would also give the comparison on outcomes built from neither edge type the sample it lacked here, since contract activity on GMX V2 fell after 2025.
- Coupling beyond a fixed transition-level mixture. Only fixed values of λ with uniform teleportation on layers weighted by US-dollar amounts, with each allowance capped, were tried (Section 3.6.3). Node-dependent or learned weights, per-layer damping, mixtures of the two stationary vectors and layers weighted as in EndorseRank and AWP remain untested, and ranking methods for multilayer networks offer designs for several of them (De Domenico et al., 2015). Any such design should keep the auditability that separates graph propagation from learned credit models, which give up interpretability to predict labelled defaults (Kotb, 2023).
- A wider token list under the same rule. The revised analysis keeps the five tokens with the most rows among those that pass the feed rule and have a daily price; they hold most of the rows, and the approvals and transfers of every other token are left out (Section 3.5.1; Table 8.4). Widening the list to every token that passes the same rule, fixed before any score is computed, would show whether the results depend on these five tokens, and it would bring back into the cohort contracts that are approved mainly on other tokens. Tokens without a qualifying feed would need another check of their prices before they could enter. Native ether moves without `Transfer` logs and enters the graphs only through WETH; adding it would need the values of transactions and internal calls, which this study did not extract.

6.4 Closing reflection

On a public ledger, an owner's decision to let a contract spend its tokens is recorded as plainly as a payment, and the two records describe different relations. A transfer records economic flow, and an allowance records delegated authority. PageRank will rank either graph, yet one ranking does not serve both constructs, and a walk over both graphs gained little over the better layer on one label and lost on the other. For a protocol or researcher that needs an auditable signal of which contracts owners entrust with their tokens, the allowance edge is the one to use on data like those studied here, the allowances of widely traded tokens valued in US dollars. On these data the count of approving owners ranked later approvals better than the walk with uniform restarts, so a propagated version has yet to show, in a registered test, that it improves on that count. For describing transfer hubs and anticipating their inflow, the transfer walk did improve on its count. None of these orderings says that a contract is safe, and a closed farm of owner addresses can raise a contract's standing at little cost (Section 5.6). Whether the authority that owners delegate to a contract tells anything about outcomes that neither graph records, such as a trader's liquidations or a contract's later misuse, remains open. The comparison registered for the few contract traders available found no lead in the registered analysis, nor when its rules were applied to the revised scores as a re-analysis, and settling the question will take more traders or labels of contract risk.